Advanced Computational Engineering
Assignment-1
SID – 530158949

1. Discrete Equations for the 2D steady diffusion problem:

Boundary conditions are given as:

(i) Temperature at upper boundary, $T = f(x) = (\sin(2\pi x/L_x))^2$

(ii) Temperature at lower boundary, $T = -f(x) = -(\sin(2\pi x/L_x))^2$

(iii) Temperature at left boundary, $T = 0°C$

(iv) Temperature at right boundary, $T = 0°C$

Spatially varying velocity field of the fluid is given as:

$U(x,y) = sin(\pi x/Lx)\, cos(\pi y/Ly) + sin(2\pi x/Lx)\, cos(2\pi y/Ly$

$V(x,y) = -cos(\pi x/Lx)\, sin(\pi y/Ly) - cos(2\pi x/Lx)\, sin(2\pi y/Ly)$

Constant values are given as: $\alpha = 80, \beta = 0.5$ and $Lx = Ly = 1$

The 2D steady diffusion equation for the fluid is given as:

$$\alpha U \frac{dT}{dx} + \alpha V \frac{dT}{dy} - \frac{d^2T}{dx^2} - \frac{d^2T}{dy^2} - \beta T = 0$$

On applying Central finite difference equation for the first and second order derivatives,

$$\frac{dT}{dx} = \frac{T_{i+1,j} - T_{i-1,j}}{2\Delta x} \quad \text{and} \quad \frac{dT}{dy} = \frac{T_{i,j+1} - T_{i,j-1}}{2\Delta y}$$

$$\frac{d^2T}{dx^2} = \frac{T_{i+1,j} - 2T_{i,j} + T_{i-1,j}}{\Delta x^2} \quad \text{and} \quad \frac{d^2T}{dy^2} = \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\Delta x^2}$$

Substituting the values for first and second derivatives and rearranging them, the equation can be written as,

$$T_{i+1,j}(2\alpha U\Delta x\Delta y^2 - 2\Delta y^2) + T_{i-1,j}(-2\alpha U\Delta x\Delta y^2 - 2\Delta y^2) + T_{i,j+1}(\alpha V\Delta y\Delta x^2 - 2\Delta x^2) + T_{i,j-1}(-\alpha V\Delta y\Delta x^2 - 2\Delta x^2) + T_{i,j}(4\Delta y^2 + 4\Delta x^2 - 2\beta\Delta x^2\Delta y^2) = 0$$

The above equation can then be written as, $AX + B = 0$; where A is diagonally dominant matrix with the coefficient values and X is matrix with Temperature variables and B is the matrix of RHS values. From the matrix A, following coefficients can be inferred as,

$a_e = (2\alpha U\Delta x\Delta y^2 - 2\Delta y^2)$; $a_w = (-2\alpha U\Delta x\Delta y^2 - 2\Delta y^2)$

$a_n = (\alpha V\Delta y\Delta x^2 - 2\Delta x^2)$; $a_s = (-\alpha V\Delta y\Delta x^2 - 2\Delta x^2)$

$a_p = (4\Delta y^2 + 4\Delta x^2 - 2\beta\Delta x^2\Delta y^2)$

Using Jacobian solver, temperature value at k[th] iteration can be found by using below formula,

$$T_{i,j}^{k+1} = \frac{-b_{i,j} - a_w T_{i-1,j}^k - a_e T_{i+1,j}^k - a_s T_{i,j-1}^k - a_n T_{i,j+1}^k}{a_p}$$

Residual for the temperature domain can be calculated as,

$$r_{i,j}^k = -b_{i,j} - a_w T_{i-1,j}^k - a_e T_{i+1,j}^k - a_s T_{i,j-1}^k - a_n T_{i,j+1}^k - a_p T_{i,j}^{k+1}$$

1. Solution for the temperature domain using Jacobian solver in MPI:

A solution matrix was obtained using maximum residual of 0.000001 as the stopping criterion.

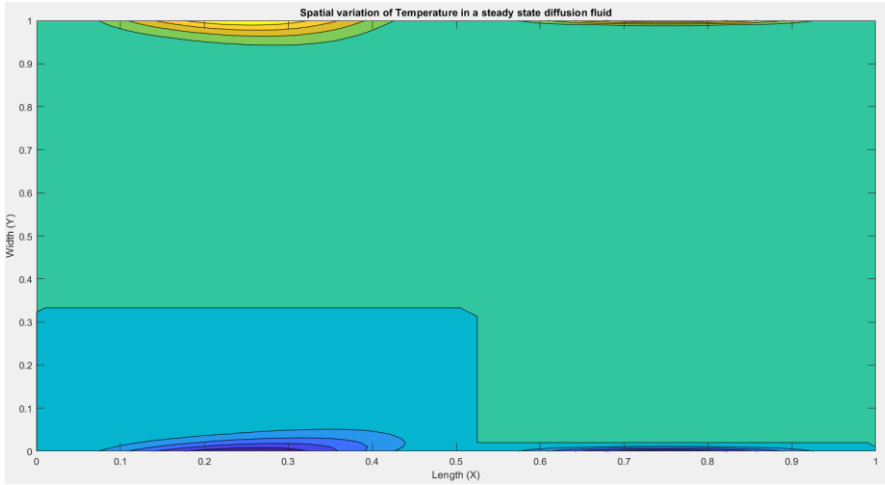

Figure 1: Contour plot for Temperature vs Spatial values

From the above contour plot, it can be observed that, maximum temperature is towards the upper boundary of fluid (yellow contour lines). Minimum temperature is found towards the lower boundary of the fluid. I can be seen that, based on closely spaced contours, the rise in temperature is steep at both upper and lower boundaries, however in the middle, the temperature is not changing. The accuracy of the solution can be seen from the below plot, as the residual is seen as converging towards the maximum value and it converges at 32nd iteration, which gives the solution for Temperature domain.

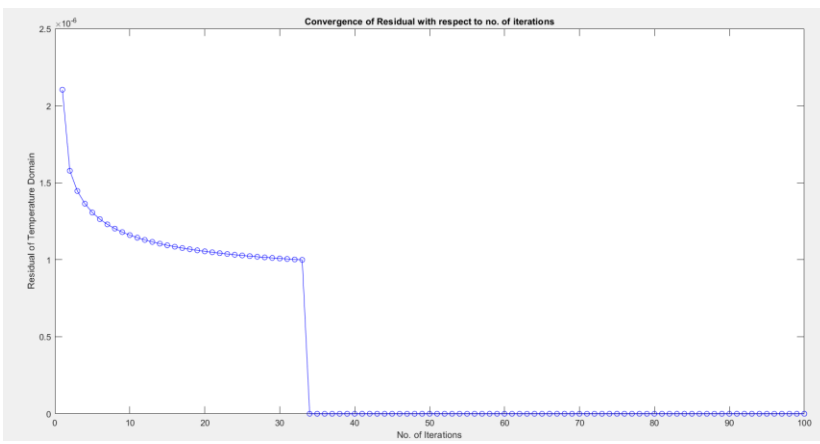

Figure 2: Residual vs No. of Iterations

2. Effect of grid size change:

The temperature domain of the solution matrix has been plotted for different grid sizes with 2 cores (np=2) and has been shown below.

No. of Cores	Grid Size	Total completion time	Maximum Temperature	No. of Iterations	Parallel Efficiency
2	1h	0.023319	0.99974827	32	0.05405
2	2h	0.024612	0.99974827	31	0.04684
2	3h	0.017474	0.99974827	26	0.05116
2	4h	0.014802	0.99974827	21	0.0506

Table 1: Variation w.r.t change in grid size (2 cores)

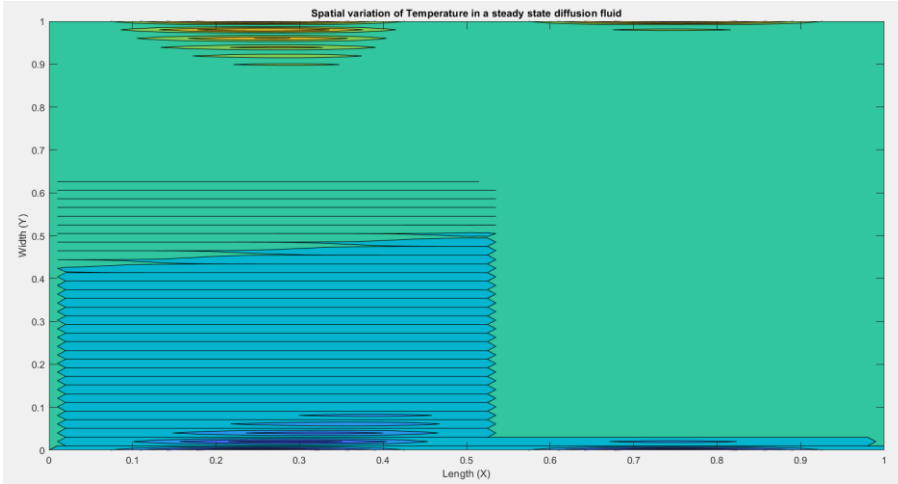

Figure 3: Contour plot for Temperature vs Spatial values (2h grid size)

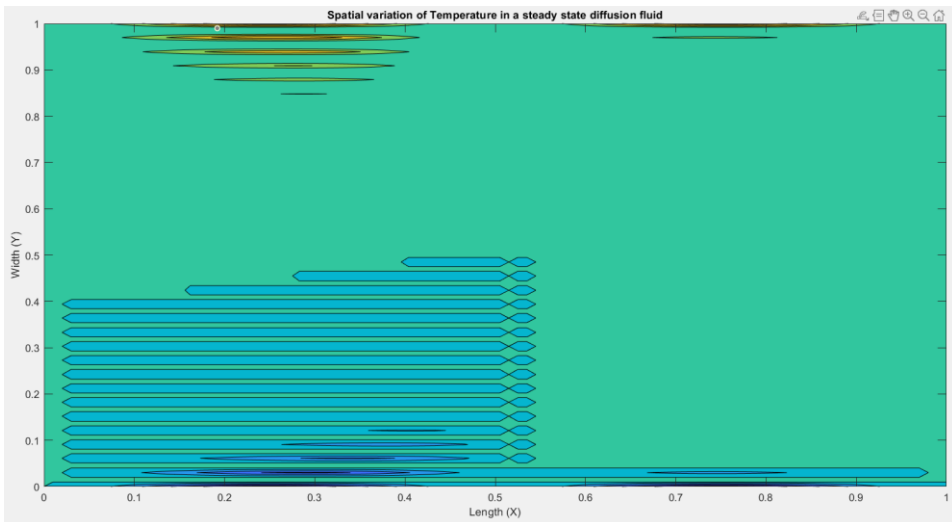

Figure 4: Contour plot for Temperature vs Spatial values (3h grid size)

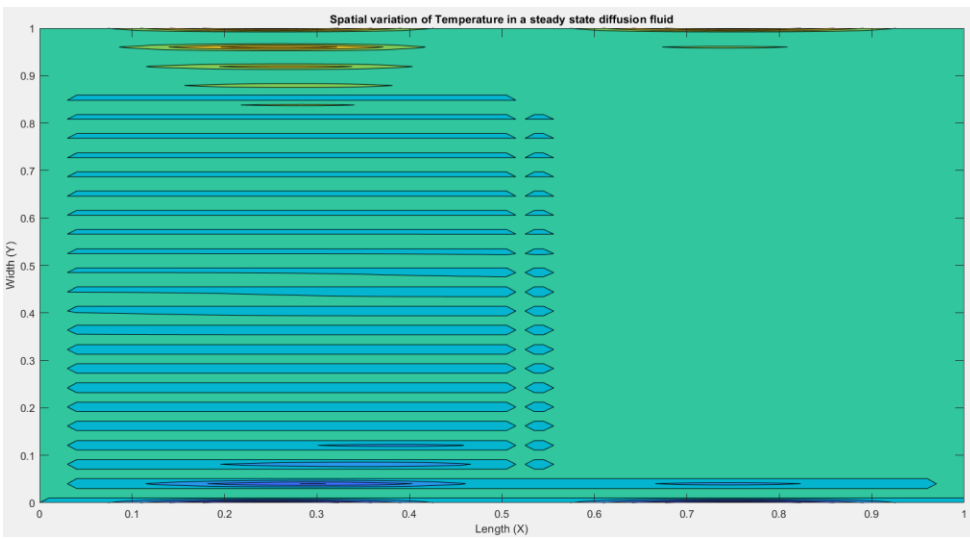

Figure 5: Contour plot for Temperature vs Spatial values (4h grid size)

It can be observed from above data, with increase in grid size, convergence occurs faster (less no. of iterations), however the temperature slope is becoming more and more shallow. The lower frequency nodes are taking longer time to converge. For a better and more accurate solution, grid sizes should not be coarse.

3. Effect of change in no. of processor cores:

No. of Cores	Grid Size	Total completion time	Maximum Temperature	No. of Iterations	Parallel Efficiency
3	1h	0.112697	0.99974827	>100	0.0074
3	2h	0.090007	0.99974827	>100	0.0085
3	3h	0.09697	0.99974827	>100	0.0161
3	4h	0.094552	0.99974827	>100	0.0053

Table 2: Variation w.r.t change in grid size (3 cores)

No. of Cores	Grid Size	Total completion time	Maximum Temperature	No. of Iterations	Parallel Efficiency
4	1h	0.11614	0.99974827	>100	0.00543
4	2h	0.115523	0.99974827	>100	0.00499
4	3h	0.109192	0.99974827	>100	0.00409
4	4h	0.125931	0.99974827	>100	0.00297

Table 3: Variation w.r.t change in grid size (4 cores)

From the above data with change in grid sizes, the number of iterations and the time taken to find the solution increases. This increases and this is due to the increase in the number of MPI operations. However, for a given number of cores, with an increase in grid size, the time taken for completion decreases. Generally, with an increase in the number of cores, the time taken for completion reduces to some extent but mostly it depends on the type of complexity of the problem. solution. Taking into account, the completion time with various grid sizes as: 1h: 0.002521sec; 2h: 0.002306sec;3h:0.001788sec and 4h: 0.001498sec. The parallel efficiency has been calculated based on these values. Efficiency is decreasing for a given grid size with increase in processor, as this again depends on the nature of the problem always and how feasible the problem is with MPI.